PROJECT BASED LEARNING REPORT

on

GestVox : Voice + Gesture Robot

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Abstract

GestVox is a multimodal human–robot interaction prototype in which the *voice* says how far to move ("three") and a *hand gesture* seen by a webcam says which way (swipe left/right/up/down). A simulated robot on a 9×9 grid executes the command. Signal-processing chains for voice, surface EMG and camera video feed SVM, random-forest and k-NN classifiers, and a *confidence gate* rejects the command when either input is unsure. Voice recognition reaches 93.6 % accuracy, the joint command 78.0 %, and a live web demo reports measured statistics for every run.

1 Introduction and Objectives

Voice and gesture are the two most natural ways to command a robot, but each fails differently: speech in noise, gestures under poor lighting or occlusion. Early systems showed that combining them is more powerful than either alone [1], and multimodal interfaces can be more robust because one channel compensates for the other [2]. In GestVox the two channels fill *different slots of one command* instead of voting on the same label: voice $\rightarrow$ number of cells $N \in \{1, \ldots, 5\}$, gesture $\rightarrow$ direction $d \in \{\text{left, right, up, down}\}$. A wrong robot move is costlier than a repeated command, so a confidence gate rejects uncertain commands.

Objectives: (i) use public datasets [3], [4]; (ii) implement at least two signal-processing techniques; (iii) compare at least two ML algorithms with accuracy, precision, recall and F1-score; (iv) propose and evaluate confidence-gated complementary fusion; (v) build a working live prototype.

2 Theory and Literature Survey

2.1 Speech and EMG signal processing

Speech is analysed in 20–30 ms frames. A pre-emphasis filter flattens its spectral tilt,

$$y[n] = x[n] - \alpha x[n-1], \quad \alpha = 0.97, \tag{2.1}$$

and mel-frequency cepstral coefficients (MFCC) [5] summarise the spectrum on the perceptual mel scale,

$$m = 2595 \log_{10}(1 + f/700), \qquad c_k = \sum_{m=1}^{M} \log E_m \cos\Big[\tfrac{\pi k}{M}\big(m - \tfrac{1}{2}\big)\Big], \tag{2.2}$$

where E_m are mel filterbank energies and f is in Hz. Surface EMG measures forearm muscle activity (useful band 10–500 Hz, 50 Hz mains interference); filtering followed by time-domain features (MAV, RMS, waveform length) and spectral features (mean frequency) is standard [6], [7].

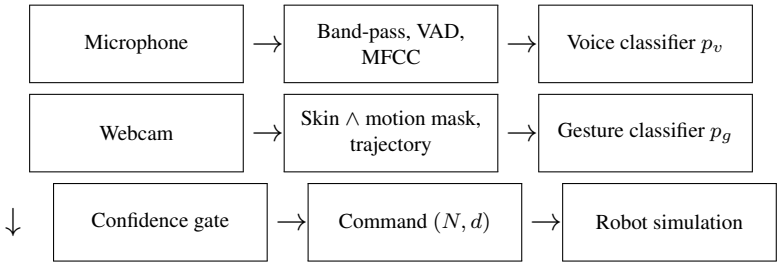


Fig. 3.1 Block diagram of GestVox.

Table 3.1 Datasets.

Modality	Dataset and use	Source
Voice	Google Speech Commands v0.02 [3]: words *one–five*, 4 000 clips, 3 000 train / 1 000 test	`download.tensorflow.org`
EMG	UCI EMG data for gestures [4]: 8 channels, 200 Hz, 36 subjects; 4 gestures → directions; subjects 1–26 train (1 831 windows), 27–36 test (686)	`archive.ics.uci.edu/dataset/481`
Camera	Live OpenCV clips; initial model trained on 600 synthetic hand clips, user clips can be added	generated by the project

2.2 Vision, classifiers and fusion

Gesture recognition extracts the hand, tracks it and classifies its motion [6]. A Haar-cascade face detector [8] removes the face; YCrCb thresholding gives a skin mask [9]; contours, convex hull and frame differencing come from OpenCV [10]. The classifiers compared are the support vector machine (RBF kernel) [11], random forest [12] and k-nearest neighbours [13], used through scikit-learn [14]. Multimodal fusion can be early, late or hybrid [15], and a classifier that may *reject* uncertain samples trades coverage for lower error among accepted ones [16]; GestVox applies this reject option to a two-modality command.

3 System Working and Methodology

3.1 Architecture and datasets

Fig. 3.1 shows the system. Table 3.1 lists the public datasets used.

3.2 Signal-processing chains

Voice: pre-emphasis (2.1), zero-phase Butterworth band-pass 80–4000 Hz, energy-based voice-activity detection, then 13 MFCCs (2.2) with deltas, cepstral mean normalisation and pooling over four time segments (130 features). **EMG:** Butterworth band-pass 10–90 Hz and 50 Hz notch, 1 s windows, MAV, RMS, waveform length, zero crossings, slope-sign changes and Welch mean/median frequency per channel (64 features). **Camera:** face masking, YCrCb skin mask with morphology, and a frame-difference motion mask so that *hand* = *skin* ∧ *moving*, which removes the static face and arm; the centroid trajectory gives net displacement $(\Delta x, \Delta y)$, path length and motion energy (10 features).

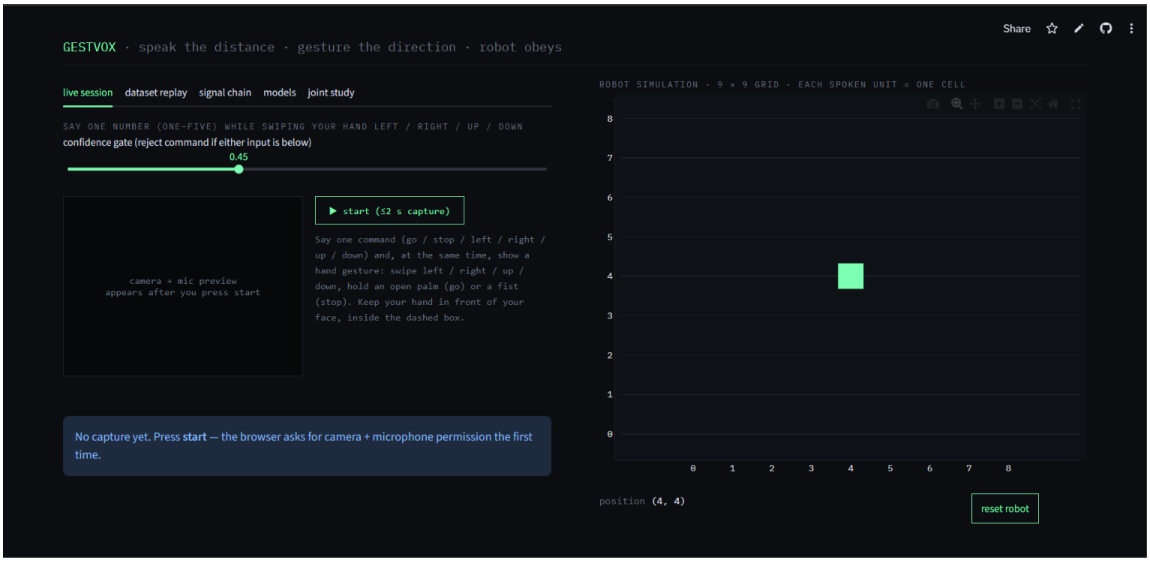


Fig. 4.1 GestVox interface: controls and measured statistics (left), robot simulation (right).

3.3 Models and confidence-gated fusion

SVM (RBF, $C{=}10$, calibrated), random forest (300 trees) and k-NN ($k{=}7$) are trained for each modality behind a standard scaler. With posteriors p_v and p_g, the command is accepted only if speech and a hand were detected and

$$\max_i p_v(i) \geq \tau \;\text{ and }\; \max_j p_g(j) \geq \tau \;\Rightarrow\; (N, d) = (\arg\max p_v,\; \arg\max p_g), \tag{3.1}$$

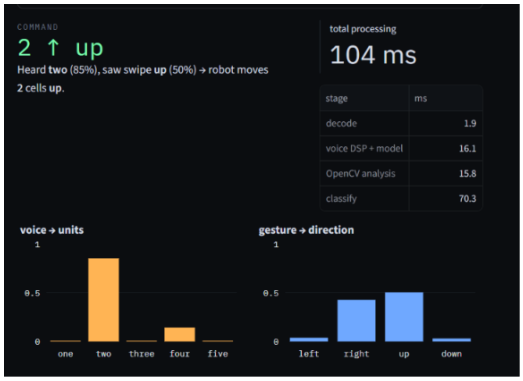
otherwise the robot does not move ($\tau = 0.45$ by default). A joint command is correct only if both slots are correct.

4 Implementation and Prototype

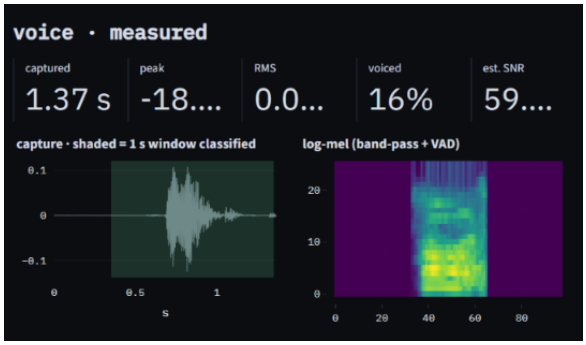
The software is written in Python 3.12 with NumPy and SciPy [17] for filtering, scikit-learn [14] for models, OpenCV 4.11 [10] for vision, and Streamlit [18] with Plotly for the web interface; only a laptop webcam and microphone are needed. A small browser component asks for camera and microphone permission and records for at most 2 s, stopping automatically 0.45 s after the spoken word ends (never before 1.2 s, so the swipe is captured). Audio (16 kHz) and frames (10 Hz, 160×120) go to Python, which shows *measured* statistics (level, SNR, hand-detection rate, trajectory, probabilities, stage times; about 0.2–0.4 s in total) and moves a 9×9 robot one cell at a time (Figs. 4.1 and 4.2).

5 Results and Discussion

Table 5.1 compares the models on held-out data (macro-averaged; EMG split is subject-independent; camera clips are synthetic, hence optimistic for real hands). The SVM is best for voice (93.6 %), the random forest for EMG (81.3 %). Removing the voice filtering chain lowers accuracy from 93.6 to 87.2 % (SVM), 91.6 to 82.6 % (RF) and 90.9 to 81.0 % (k-NN), so the DSP chain adds 6–10 points; on EMG filtering changed accuracy by only about ± 2 points.



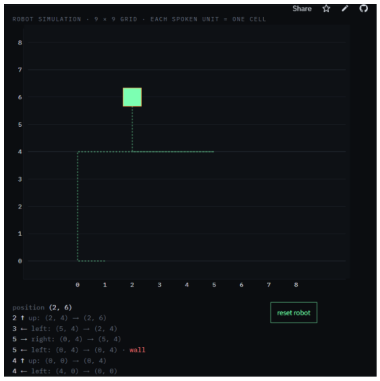
(a) result: units, direction, probabilities



(b) OpenCV: contour, hull, skin mask



(c) voice: waveform, classified window



(d) robot after a multi-cell move

Fig. 4.2 Prototype screens: (a) command result, (b) gesture analysis, (c) voice analysis, (d) robot simulation.

Table 5.1 Test accuracy / F1-score of the compared models.

Model	Voice (5 cls)	EMG (4 cls)	Camera (4 cls)
SVM (RBF)	**0.936** / 0.937	0.789 / 0.787	0.967 / 0.967
Random forest	0.916 / 0.916	**0.813** / 0.813	**0.980** / 0.980
k-NN	0.909 / 0.909	0.774 / 0.773	**0.980** / 0.980

Table 5.2 Joint command accuracy (%) versus SNR with noise on one modality.

Noise on	Clean	20 dB	10 dB	5 dB	0 dB
Voice	78.0	75.7	53.7	28.1	19.4
EMG	78.0	76.7	59.3	37.6	26.9

Joint accuracy (units *and* direction correct, random voice–EMG pairing) is 78.0 % on clean data and falls with noise (Table 5.2). The gate trades coverage for accuracy: at $\tau = 0.5$ it executes 94.1 % of commands at 80.4 % accuracy; at $\tau = 0.7$ it executes 62.1 % at 88.7 % accuracy and avoids 68 % of the wrong moves (Fig. 5.1).

Because a complementary command needs both slots, its accuracy is limited by the weaker modality; the gate converts many errors into harmless rejections. In live tests voice worked

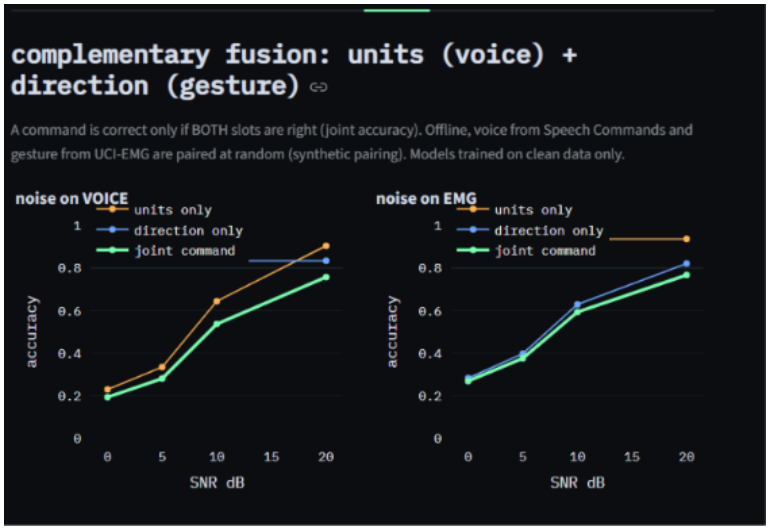


Fig. 5.1 Joint study: accuracy under noise and gate-threshold sweep.

reliably, while the camera gesture depended on lighting and a clear, fast swipe.

6 Conclusion and Future Work

GestVox shows that splitting a command between voice (distance) and gesture (direction), with a confidence gate, gives a simple and explainable multimodal interface; signal processing was essential for voice, three classifiers were compared on public datasets, and a live prototype reports measured statistics. Learning outcomes: filter and feature design, classical computer vision, fair model comparison, fusion with a reject option and real-time web deployment. Touch-free control is relevant to assistive devices, collaborative and warehouse robots, sterile environments and smart homes. Limitations: simulated robot, random voice–EMG pairing, a camera classifier initialised on synthetic hands and lighting dependence. Future work: learned hand landmarks, a paired voice–gesture dataset, a larger vocabulary and a physical robot.

References

[1] R. A. Bolt, “Put-that-there: Voice and gesture at the graphics interface,” *ACM SIGGRAPH Comput. Graph.*, vol. 14, no. 3, pp. 262–270, Jul. 1980.

[2] S. Oviatt, “Ten myths of multimodal interaction,” *Commun. ACM*, vol. 42, no. 11, pp. 74–81, Nov. 1999.

[3] P. Warden, “Speech commands: A dataset for limited-vocabulary speech recognition,” arXiv:1804.03209, Apr. 2018. [Online]. Available: https://arxiv.org/abs/1804.03209. [Accessed Nov. 12, 2026].